



SECURE WINDOWS USING OPEN SOURCE SOFTWARE

In general, when one uses the term ‘open source’, Linux is probably the first thing that comes to mind. But even if you’re a Windows user, you can use open source software! This article shows you how, by touching upon two key security measures: encryption and anti-malware. So without further ado, let’s get started!

You don’t have to be part of the FBI to use this type of software, but you can be as secure. Even if your hard disk or flash drive gets stolen, encrypting your data will save the day. After all, the hardware is expendable, your data isn’t. One of the best encryption software out there is good ol’ TrueCrypt—it provides the most robust algorithms to encrypt not only bits of data, but full hard drives as well. The key benefit of something like TrueCrypt is that it provides on-the-fly encryption, and it is cross-platform—offering encryption benefits across Windows, Linux and Mac systems. What this really means is that your data will remain encrypted even when you’re using it. The system will transparently encrypt or decrypt your files in the background, as required. I will show you how to encrypt your data in two modes (make sure you back up your data before attempting to encrypt anything—something as simple as a power-cut during the initial encryption process can cause data loss that may be irrecoverable, especially while encrypting drives or partitions).

1. **Containers:** This is done by creating an encrypted file, known as a ‘container’. This file is then mounted in Windows as a drive. Data within this drive is encrypted in real-time, as explained above.
2. **Full-disk encryption:** This mode is safer than the first. As the term implies, it encrypts your entire drive or partition. Before we begin, let’s download and install TrueCrypt (from the link given at the end of the article) and run the set up.

Container-based encryption

Containers are an easy starting point if you’re new to (and apprehensive about) encryption. It’s a pretty straightforward process. Start TrueCrypt from the *Start Menu*. A screen similar to Figure 1 appears. Click the *Create Volume* button. In the first step of the *Volume Creation Wizard*, select ‘*Create an encrypted file container*’ and press *Next*. In the *Volume Type* step, select ‘*Standard TrueCrypt Volume*’—unless you want to be super-secretive and store all those evil ideas lying around on your hard disk, in which case, you can select ‘*Hidden TrueCrypt Volume*’. The latter option creates a hidden container within the main container, somewhat like a Matryoshka doll (that wooden Russian toy that is a doll, within a doll, within a doll...). Then click ‘*Next*’.

In the *Volume Location* screen, select a secure location and filename (give it any extension—even a commonly known one, to mask its true identity). As always, hit *Next* to continue. At this point (Figure 2), select the encryption algorithm to be used to encrypt your container. TrueCrypt also gives you the option of using multiple algorithms. AES is commonly used, so let’s continue with it for this example; click *Next* to proceed. At the next step, you will be asked to specify the size of the container—enter an appropriate value, and click *Next*.

An important step follows—enter the password! Once you have and clicked *Next*, you may be warned if your password is less than 20 characters. Don’t worry, just click *OK* and continue. Move your mouse randomly inside the window (Figure 3) to create a unique set of encryption keys. Once